DE.AUTO

CHAPTER 7

FORCED INDUCTION

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English

MEMBER EXCLUSIVE • DE.AUTO •

POWERED BY DE.AUTO

DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE
MUST-KNOW TERMS

Car modifications, explained

Chapter Seven: Turbochargers and Superchargers

“BREATHE IN. BOOST UP.”

— DE.AUTO

Important notice

This guide is **entertainment only**. It is not educational or scientific content, and it is not a how-to manual. DE.AUTO is not an accredited scientific, engineering or technical authority. Everything here is simplified for fun and general interest, and it can differ from one car to another.

You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

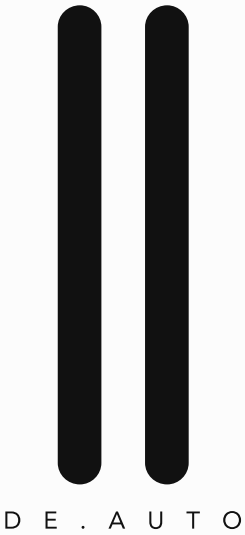
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

DE.AUTO — Because knowledge is power, and cars are passion.

READY?

A small engine that rivals a bigger one? 🌀 What makes boost work in your favour?

By the end of this chapter you'll know:

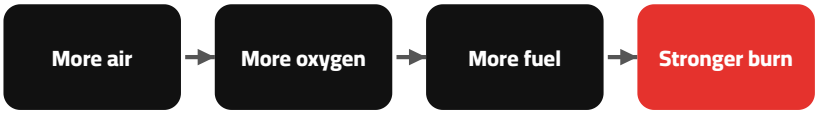
- ✓ Why more air = more power, and what oxygen has to do with it
- ✓ Turbo anatomy: compressor, turbine, wastegate, blow-off, intercooler
- ✓ What boost is, and why bar and psi aren't horsepower
- ✓ Why the delay happens (Turbo Lag) and how it's fixed
- ✓ What the "tssss" from some cars is, and what flutter means
- ✓ Superchargers: Roots, twin-screw, centrifugal vs. turbo
- ✓ Switch off straight after driving? Plus oil and wear signs



Why compress the air at all? 💨

A normal engine “breathes” by itself: the piston goes down and sucks the air in. This is called **natural breathing** (Naturally Aspirated). As for **forced induction** it means: we don’t wait for the engine to suck the air in, we **push the air in under higher pressure**.

The chain starts with one thing: how much air enters?



Air isn’t power: the engine burns fuel, and fuel needs enough oxygen.

Same volume, different number of air molecules

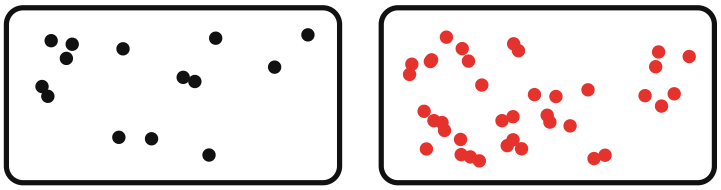


Illustration: each box is the volume of one cylinder, and the dots are air molecules

21%

That is roughly the share of oxygen in air. The rest is mostly nitrogen, which takes no part in combustion. So the more air you squeeze into the same volume, the more oxygen gets in.

✕ MYTH

“A small turbocharged engine gives free power and always saves fuel”.

✓ FACT

Nothing is free. A turbo lets a small engine work more efficiently in gentle driving, but when you ask for high power it drinks more fuel because it burns more mixture. The saving depends on how you drive.

🎯 Challenge 1 **What is the main thing that increases when we compress the air inside the engine?**

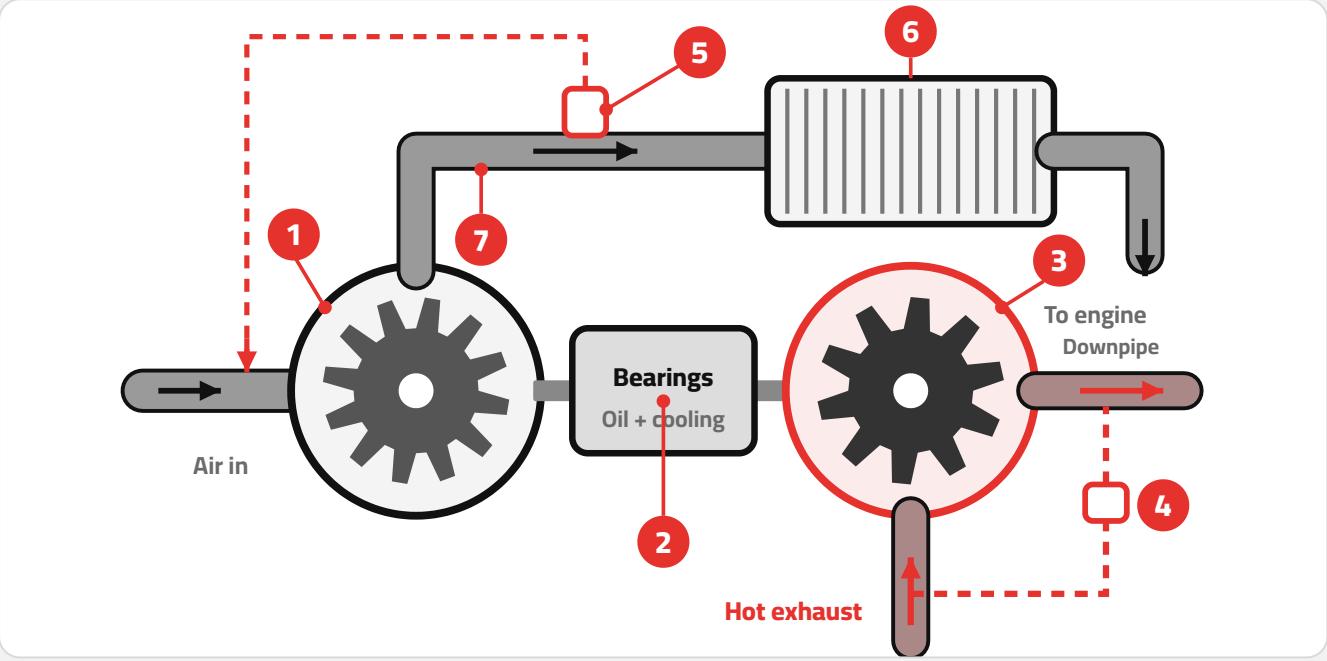
A) The amount of oxygen in the same volume

B) The number of cylinders

C) The colour of the flame

Anatomy of a turbo: two fans on one shaft 🌀

The turbocharger, in short, is **two fans joined on one shaft**. One is spun by hot exhaust gas, and the other (on the same shaft) compresses clean air and sends it to the engine. So it uses energy that was going to waste out of the muffler!



Simplified illustration: air flows in from the left, and the exhaust from below. The shapes and positions of the parts differ from car to car

1. Compressor
The “cold” fan: it sucks in air and compresses it.

2. Shaft and bearings
Joins the two fans and spins on a film of oil.

3. Turbine
The “hot” fan: exhaust pushes it round. In some small turbos its speed exceeds 200,000 rpm!

4. Wastegate
A valve that diverts part of the exhaust away from the turbine so the boost doesn’t go over the limit.

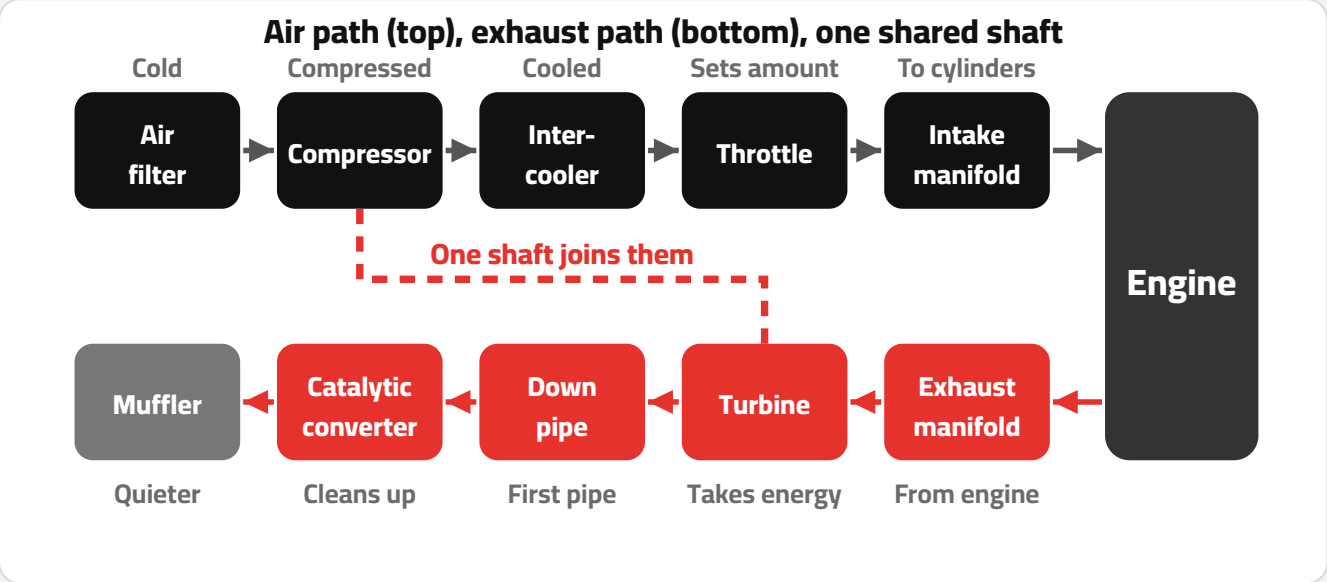
5. Blow-off / bypass valve
A valve that releases trapped pressure when you lift off the throttle.

6. Intercooler
A radiator that cools the compressed air before it reaches the engine.

7. Boost pipes
Carry the compressed air, and any leak in them wastes boost.

The complete air journey 🗺️

Everything in a turbo system runs along two connected paths: the air path (cold, then compressed) and the exhaust path (hot). Along the way there are “checkpoints”:



Simplified. Some cars combine the cat with the downpipe or change the order slightly

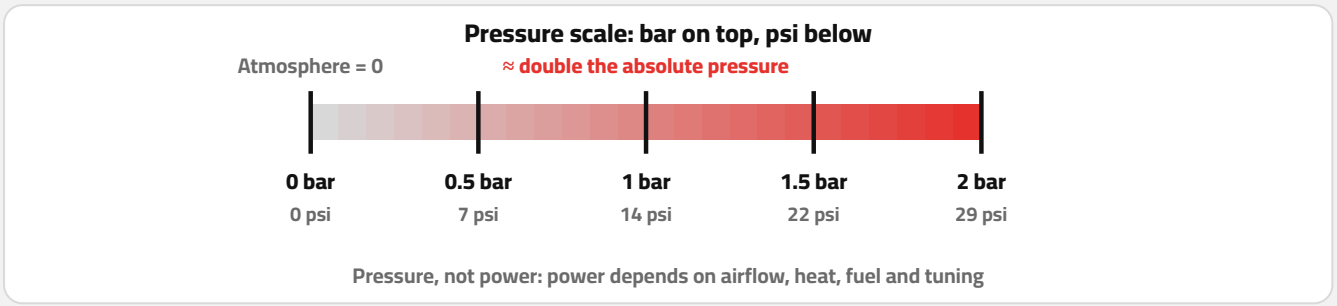
1. Air enters through the **air filter** and passes sensors that measure how much there is. (See **Chapter 3**)
2. **The compressor** squeezes it, and it heats up from the compression itself.
3. **The intercooler** cools it, so its density rises and the risk of knock falls.
4. **The throttle** controls how much air gets in according to your pedal, and the **manifold** distributes it to the cylinders.
5. The computer adds the right amount of fuel, and combustion happens.
6. The exhaust leaves through the **exhaust manifold** and pushes the **turbine**, which spins the compressor. Then it flows through the **downpipe** to the cat and the muffler. (See **Chapter 4**)

> 🔁 **The idea:** There is no “free pressure” from ordinary intake. The compressor does the work, and the turbine gives it the energy. And the harder you press the pedal, the higher the exhaust temperature and the rotation speed.

✅ **DE.AUTO tip:** If a pipe or a joint leaks pressurised air, you may hear a light whistle along with a drop in response. Don’t put off the check: a leak makes the turbo work harder to compensate.

Boost: bar and psi, and what they mean 📏

Boost is the extra pressure the compressor creates **above atmospheric pressure**. You hear it in two units:



The numbers are for illustration only: each car has its own boost depending on its engine design

Bar

The metric unit. Atmospheric pressure at sea level is roughly 1 bar.

Pounds per square inch (psi)

The American unit. Every 1 bar ≈ 14.5 psi.

≈2×

A boost of 1 bar means the absolute pressure in the intake pipes is about double atmospheric. In theory, at the same temperature, that is close to double the amount of air in the same volume, but heat and efficiency make reality lower. And the final power doesn't double by the same proportion.

The boost gauge in cars usually reads the pressure difference from the atmosphere (Gauge), so it reads 0 when you're standing still.

The MAP sensor (see **Chapter 3**) measures absolute pressure, and from it the computer knows how much boost is going in.

Many factory cars run on moderate boost, and some designs go higher. The exact figure is set by the manufacturer.

✕ MYTH

"The more boost, the more power, in the same proportion".

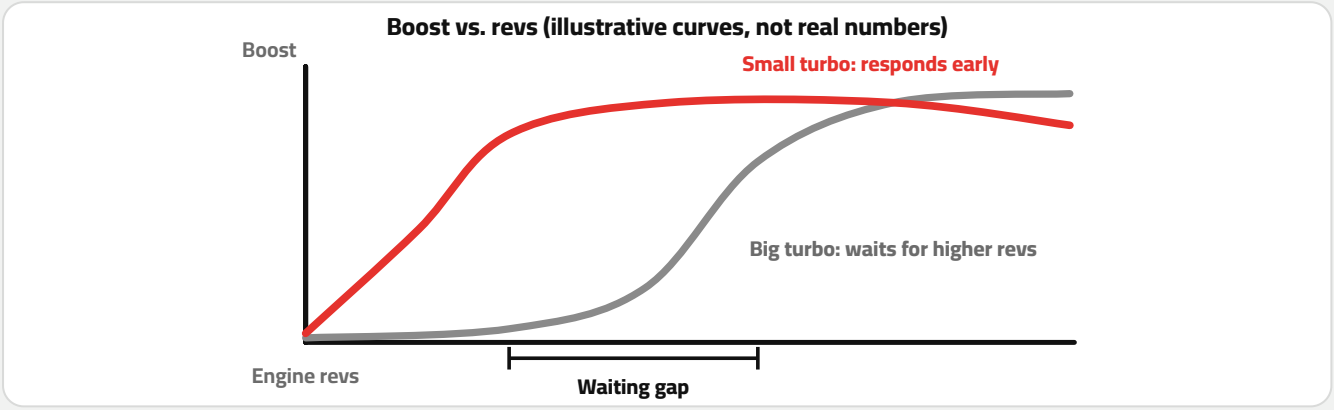
✓ FACT

Boost is pressure, not power. Hotter air means lower density, and an ill-suited turbo heats the air more. And every increase in boost adds heat load and stress on the pistons, the gasket and the oil. That is why the manufacturer picks a safe figure for its engine.

☀️ **For hot weather:** Your altitude and the air temperature change the air's density. In a hot summer the incoming air is less dense, and the intercooler has a harder job. That is why cooling matters especially here!

Turbo lag ⏳

You press the pedal… and nothing happens at that moment? That is the lag. The reason is simple: the turbine needs **enough exhaust** to spin it, and the exhaust needs an engine that is turning and loaded. At low revs there is little exhaust, so the compressor doesn't squeeze enough, until the turbo "wakes up".



How did they solve it?

Small turbo
Light and quick to spin, so it reaches boost early. But the air may get tight for it at high revs.

Twin-scroll
The turbine has two inlets that keep the pulses of different cylinders from overlapping. The exhaust arrives cleaner, so it spins faster.

Variable geometry (VGT)
Vanes that move to narrow and widen the turbine inlet depending on revs. Common in diesels, and less so in petrol engines because of the high exhaust temperature.

Twin turbo
Two in parallel (one per bank) or in sequence: a small one for low revs and a big one for high revs.

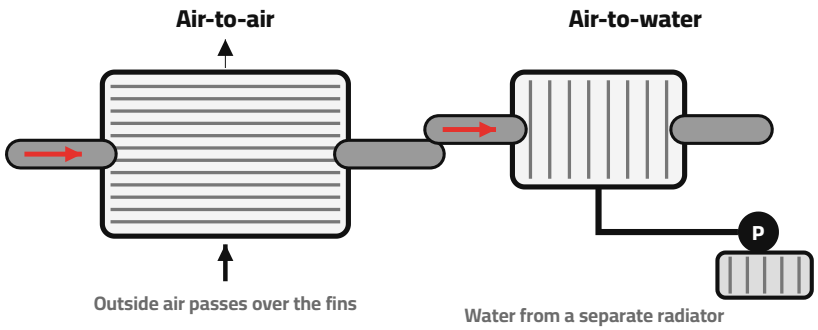
Electric assist
An electric motor spins the compressor or helps the turbo at the start, filling the gap.

Ball bearings
Reduce shaft friction, so the turbo responds faster.

⚖️ **The trade-off:** Manufacturers are always balancing: early response against higher breathing at the top end of the revs. No solution beats everything at everything, and the good part is you understand why your car is set up the way it is.

Heat and the intercooler 🔥❄️

When you compress air, it heats up. That is physics, not a flaw in the turbo. Think of a bike pump getting hot when you pump fast. And here is the problem: hot air is **less dense** (less oxygen in the same volume) and **closer to knock**.



Air-to-air: cooled by outside air. Air-to-water: cooled by water from a dedicated cooling loop, and it shortens the air pipes

Air-to-air

The simplest and most common. It needs a spot facing the oncoming airflow, and slightly longer pipes.

Air-to-water

Compact, shortens the air path and improves response, but adds a pump, a radiator and extra parts.

🌡️ **Heat build-up:** After hard driving and then standing in traffic, the intercooler holds on to heat (Heat Soak), so the computer pulls timing to protect the engine and the car feels a little weaker. It's normal, and it comes back once it cools down.

Compression and knock with boost

A turbo engine compresses the air twice: once with the compressor and once with the piston, so pressure and temperature inside the chamber are higher. And if the mixture ignites by itself before its time, that is **knock**, and it damages the engine. That is why:

Many turbo engines use a slightly lower **compression ratio** than naturally aspirated ones. Direct injection and smart cooling have narrowed the gap over time.

The computer pulls back the ignition timing immediately if it senses knock.

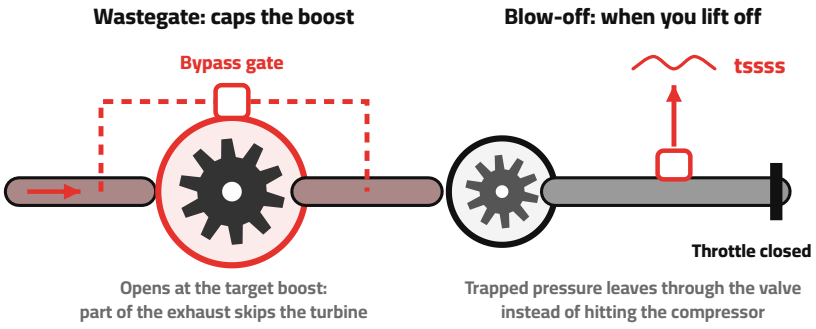
Octane matters: the more boost and heat, the more you need a fuel that resists knock better. (See **Chapter 3**)

🎯 Challenge 2 **What does the intercooler do?**

A) Cools the compressed air | B) Spins the turbine faster | C) Releases the excess pressure

Wastegate vs. blow-off: who makes the noise? 🔊

The two that people confuse most. Each has a completely different job, in a different place:



Simplified illustration. Both are on the turbo, but their positions and designs vary

Wastegate

On the **exhaust side**. Sets the maximum boost by controlling how much exhaust spins the turbine. Without it, boost rises above the limit.

Blow-off / bypass valve

On the **air side**. Opens the moment you lift off the throttle so the trapped pressure is released instead of slamming into the compressor.

What is the “tssss” and what is the “flutter”?

The “tssss” (Blow-off): the valve vents the pressure to the atmosphere, and you hear a quick hiss.

The flutter (Compressor Surge): if the valve doesn’t open, the trapped pressure comes back and hits the compressor wheel while it’s spinning, and you hear “fl-fl-fl”. Repeated, it wears the bearings.

Most factory valves **recirculate the air** back to the intake (Recirculating), so they are quiet or you don’t hear them at all.

✕ MYTH

“A blow-off valve gives you power”.

✓ FACT

The valve protects the compressor and improves response after gear changes, and it is more about sound than performance. And venting to atmosphere in some cars can upset the computer’s air calculation. And mind this: there are limits on loud noise in regulations.

🎯 Challenge 3 **If you hear a “tssss” right after the driver lifts off the throttle, which part most likely made the sound?**

A) The wastegate | B) The blow-off valve | C) The intercooler

Oil, cooling and signs of wear 🛢️

The turbo works in a harsh environment: a very hot side, a cold side, and bearings spinning at incredible speed on a thin film of oil. They are protected by **oil** and **coolant**.

Oil
Feeds and cools the bearings. Old, wrong or low oil = the most common cause of turbo failure. Stick to the type and change interval in your car's manual.

Coolant
Many modern turbos have a water passage cooling the housing, and some cars keep pumping coolant after the engine is switched off.

Everyone's question: switch off straight away, or let it idle a bit? 🤔

Normal driving: nothing special needed before switching off.

After hard, long or hot driving (climbing a mountain, towing, or a very hot day): ease off for the last minute or so, or let it idle gently for half a minute to a minute before switching off, so the bearings cool down.

Don't do a "final stab on the pedal" before switching off. The strongest reference is always: **your car's manual**.

✕ MYTH
"A turbo ruins the engine and shortens its life".

✓ FACT
A factory turbo is designed with its engine and lasts for years if looked after with oil and maintenance. Failures usually come from neglected oil, unsuitable parts, or raising boost without calculations.

Signs of trouble: when to worry? 🚨

Blue or grey smoke
May indicate oil leaking into the air or exhaust side (a technician determines the cause).

A high, sharp whine or grinding
Far louder than normal, which could mean tired bearings or a damaged blade.

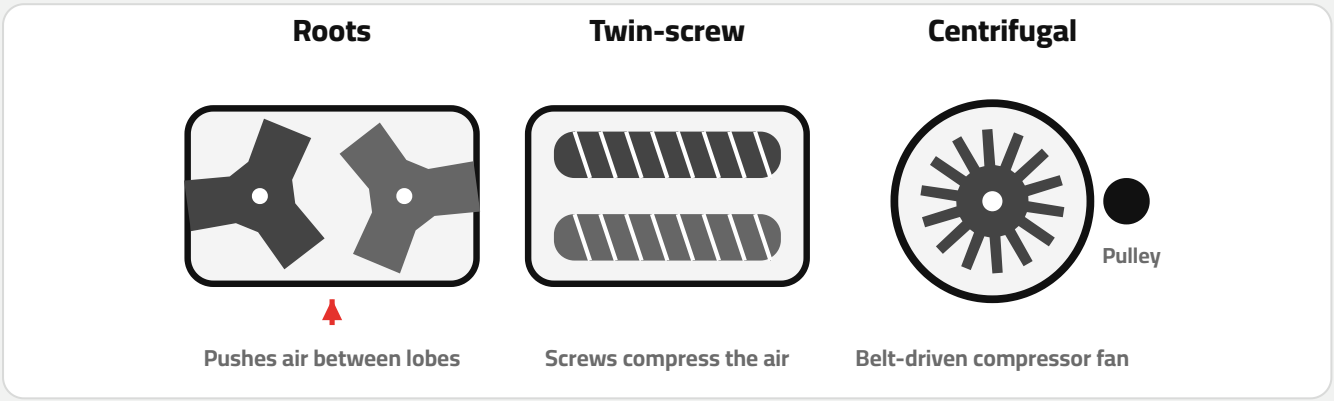
Sudden weakness with the engine light
The car may enter "protection mode" (Limp Mode) and limit the power.

Excess oil use or a boost leak
Oil that needs topping up between services, or a leaking whistle with weak response.

🔍 **The rule:** Most failures are caused by **oil** or a **foreign object** getting into the air. Any sign above: go see a technician. (Maintenance is in **Chapter 14**)

The supercharger: the belt-driven blower 🔄

The supercharger does the same job as the compressor: it squeezes the air. The key difference: it doesn't take its energy from the exhaust, it takes it from **the engine itself** via a belt, chain or gears. In other words it turns with the engine immediately.



Simplified illustrations of the idea; real designs are more complex

Roots

Two lobes rotate and push air from one side to the other, with no internal compression. Boost from the first revs, and it has the well-known whine. It heats the air more, and usually wants an intercooler.

Twin-screw

Two helical screws compress the air internally. More efficient than Roots, with early boost. A distinctive sound, and more precise to manufacture.

Centrifugal

Resembles the turbo's compressor but driven by a belt. Boost rises gradually with revs, and efficiency is high. The whine rises as revs go up.

What's good and what's hard?

The good ✅

Almost instant response with no "lag", a natural and direct pedal feel, and no high exhaust heat near the compressor.

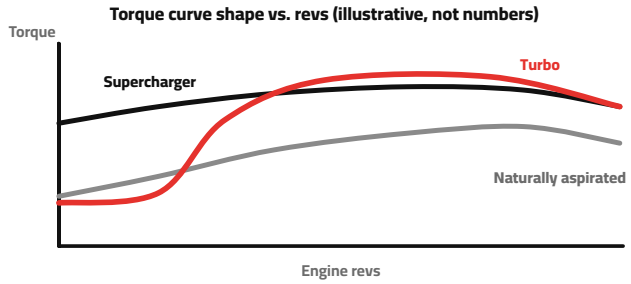
The hard ⚠️

It takes some engine power to spin it (a loss), raises the air temperature, and sometimes makes a loud whine.

🎯 Challenge 4 **What is the energy source that spins a supercharger?**

A) Exhaust gases | B) The engine, via a belt or gears | C) The car battery

Which is best? An unbiased comparison ⚔️



The shapes show the general "character"; actual figures differ from engine to engine

Comparison	Turbo	Supercharger	Naturally aspirated
Power source	Engine exhaust	Belt/gears from the engine	The engine alone
Response	Excellent once boosted, with a small delay at first	Almost instant	Instant and linear
Efficiency	High: it uses wasted energy	Lower: it takes from the engine	Simple and steady
Heat and complexity	High exhaust heat, plus oil and cooling	Air heat, maintenance cost	Simplest, fewest parts
Sound	Soft whistle + valve sounds	A clear whine that identifies the type	Pure engine sound
Typical use	Family cars, performance cars and diesels	Performance and muscle cars	Varied and lightweight cars

✅ **DE.AUTO tip:** There is no absolute "best". Choose by your use: economy? response? simplicity? maintenance? sound?

Diesel and turbo: an old story 🚛

Almost every modern diesel car has a turbo. The reason is that a diesel usually runs on excess air (it doesn't need to balance the mixture like a petrol engine), so the turbo gives it more air and raises torque noticeably.

Exhaust heat in a diesel is usually lower than in petrol, so **variable-geometry** designs are very common on diesels.

Diesel doesn't rely on octane; it has its own number called **cetane**.

The same rules apply: oil, intercooler and maintenance.

If you want to understand tuning: the full picture 🧩

A turbo isn't a part you fit and forget. It is a **system** that has to work in harmony: air, fuel, heat and tuning.

Fuel
More air needs more fuel. If fuel doesn't increase in step, the mixture goes "lean", runs hot and harms the engine.

Octane
High pressure and heat may call for a fuel with higher knock resistance. (**Chapter 3**)

Downpipe
The first pipe after the turbo. Changing it affects flow, sensors and emissions. (**Chapter 4**)

Tuning
Coordinates boost, fuel and timing. (Details in **Chapter 12**)

✕ MYTH
"A bigger turbo is always better".

✓ FACT
A bigger one squeezes more air at high revs, but it lags more at low revs and may not suit street driving. "The most suitable" for your engine and use matters more than "the biggest".

🚫 **No promises:** Turbo and boost modifications affect the warranty, emissions, insurance and the roadworthiness inspection, and must be legal and done by a certified specialist. We don't give you steps for doing it or promises of more power.

Common mistakes (don't make them!) ❌

❌ **Raising boost "just a little" without tuning that sorts out fuel and timing:** heat and knock, and it can void the warranty.

❌ **An unsuitable air filter or damaged intake pipes:** dust is the turbo's number one enemy, especially in our climate.

❌ **Ignoring the engine light or "protection mode":** those are messages, not nuisances.

🎯 Challenge 5 **Why does a boosted engine need more fuel when more air goes in?**

A) To keep the mixture ratio and cool the chamber
B) To make more noise
C) To lighten the car

Garage talk vs. the proper name 🔧

Each term in its own row, so you don't mix up the "blow-off" and the "wastegate":

Gulf garage talk	Proper name	What it means
Turbo	Turbocharger	An air charger driven by exhaust gases
Super / Supercharger	Supercharger	An air charger driven by the engine through a belt or gears
Kombrisar	Compressor Wheel	The fan that compresses the air inside the turbo
Turbeen	Turbine Wheel	The fan that the exhaust spins
Boost	Boost	The extra pressure above atmospheric
Westgate	Wastegate	A valve that sets the maximum boost by diverting some exhaust
Blow-off	Blow-off Valve (BOV)	A valve that releases pressure when you lift off the throttle, and the tssss comes from it
Bypass / diverter	Bypass / Diverter Valve	Returns the compressed air to the intake instead of venting it to the atmosphere
Interkooler	Intercooler	A radiator that cools the compressed air
Lag	Turbo Lag	The delay between the pedal and boost arriving
Twin scroll	Twin-Scroll	A turbine with two inlets that improve response
Rafrafa (flutter)	Compressor Surge	The pulse of air bouncing back onto the compressor wheel
Down pipe	Downpipe	The first pipe after the turbo
Turbo timer	Turbo Timer	A device that keeps the engine running for a minute after switching off. Unnecessary for most modern cars
Heat soak	Heat Soak	Heat building up in the intercooler after standing or hard driving

✅ **Try talking the talk:** "My turbo has an internal wastegate and a blow-off valve that recirculates to the intake, and the intercooler is air-to-water." If you can follow that sentence without mixing anything up, you are talking with the garage folks like a professional. 😎

➡️ More power... and what holds the car on the road? The suspension, the springs and the shock absorbers... and why do some cars roll in corners? It's all in **Chapter 8: Suspension and Handling**!

Wrap-up 📌

Now you know that a turbo is two fans on one shaft, that boost is pressure and not power, and that the intercooler, oil, fuel and tuning are what protect the system. You know the difference between the wastegate and the blow-off valve, and between Roots, twin-screw and centrifugal, and why lag happens. And if someone comes along and says "a bigger turbo is always better"... **you're ready to reply, because you learned it from DE.AUTO!** 🔥

🎯 Challenge answers:

1) The amount of oxygen in the same volume.

2) It cools the compressed air.

3) The blow-off valve.

4) The engine, via a belt or gears.

5) To keep the mixture ratio and cool the chamber.

➡️ Chapter 8: **Suspension and Handling**. Power without grip and stability is worth nothing, and we'll see how the car holds the road.